

## **PRACTICAL 1:**

### **Practical1(1)**

#### **Problem Phase:**

This problem is about **Left Rotation of an Array**. The array elements are shifted to the left by d positions. The first elements move to the end of the array. The main aim is to rotate the array **in-place** without using extra storage.

#### **Solution:**

Store the first element temporarily. Shift all elements one position to the left. Put the stored element at the last position. Repeat the process d times. Finally, print the rotated array.

### **Practical1(2)**

#### **Problem Phase**

This problem is about **finding duplicate elements in an array**. The program checks every element with the elements after it to find numbers that occur more than once.

#### **Solution**

The program uses **two nested loops**. The first loop selects an element, and the second loop compares it with the remaining elements. If both elements are equal, the number is printed as a duplicate. This solution does not require any extra array.

### **Practical1(3)**

#### **Problem Phase**

This problem is about **finding the longest word in a sentence** and counting the number of characters in that word. The program reads a sentence and checks each word one by one.

#### **Solution**

The program builds each word character by character until it finds a space. It then compares the word length with the current longest word. If it is longer, it becomes the new longest word. Finally, the program prints the longest word and its number of characters.